

Functional Requirements – Smart Dog Collar

Category	Requirement
Health Monitoring	- System shall measure and record dog's body temperature . - System shall measure and record heart rate .
Activity Monitoring	- System shall detect and classify activity into Resting, Walking, and Running .
Location Tracking	- System shall provide real-time GPS-based location of the dog.
Alerts & Notifications	- System shall send Email notifications to the owner in case of abnormal temperature, abnormal heart rate, or unusual activity.
Connectivity & Data	- System shall transmit data to the cloud via Wi-Fi . - System shall store and manage data on Cloud .
User Interaction	- Owner shall access data through a mobile/desktop URL . - Dashboard shall display real-time health data and location on a map.
SOS Functionality	- Collar shall include an SOS button that, when pressed, sends an email alert with the current location to the owner.
Buzzer	- System shall activate a buzzer on the collar for recall or emergency alerts.

System Flow (High-Level)

1. **Sensors** collect body temperature, heart rate, and activity (resting/walking/running).
2. **GPS** acquires real-time location.
3. **Microcontroller** processes the data.
4. Data is sent via **Wi-Fi** to the **Firestore backend**.
5. Backend triggers **email notifications** if abnormal health/activity is detected or SOS button is pressed.
6. Owner accesses real-time data (health + location) via **mobile/desktop URL**.